

Zhuoyang JIANG

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EDUCATION

The Hong Kong University of Science and Technology, China Sep.2023 – Jun.2025

MSc in Data-Centric AI Technology (DCAI) | GPA: 3.74/4.3 | Thesis Advisor: [Prof Lei Chen](#)

Southeast University, China Sep.2019 – Jun.2023

BEng in Artificial Intelligence | GPA: 3.35/4.0 | Thesis Advisor: [Prof Conghui Zhou](#)

WORK EXPERIENCE

Citymind Lab - HKUST(GZ) | *Research Assistant* Nov.2025 – (Expected)Jun.2026

- Conducting research and development of Spatial-temporal Foundation Model in Urban Computing Tasks.

Zhongguancun Academy | *Research Intern*

Aug.2025 – Nov.2025

- Conducted research on graph representation learning for autoregressive fundamental models.

Microsoft Research AI for Science | *Research Intern*

Aug.2024 – Jun.2025

- Conducted research and development of Science Foundation Model (SFM) and its Agent (Science Copilot).

PUBLICATION

- Zhou Conghui, Yang Liuyi, **Jiang Zhuoyang**, Wu Xinyu. [A Deep Learning Approach to Identify Potential Sites for Pocket Park Installation in Nanjing, China](#) [J]. Journal of Urban Technology (JCR Q1)

* Research figure selected as cover. Distribution of contributions are detailed in [“Notes on contributors”](#).

RESEARCH EXPERIENCE

Graph Representation Learning for Science Foundation Models | *AI4Science*

(Sub-projects of MSR AI4S SFM: Supervised by [Dr Yaosen Min](#) and [Dr Peiran Jin](#))

Aug. 2024 – Present

- Positioning:** A novel graph representation approach to improve molecular compression and enhance the generalization ability of Science Foundation Models (SFM) for drug discovery tasks.
- Proposed a new graph-based molecular tokenizer that enhances token compression and generalization for autoregressive models (**Data-centric**); empowered the model through general-purpose large-scale pretraining and task-aware fine-tuning for diverse molecular tasks (**Algorithm-driven**); evaluated on real-world ADMET benchmarks with SOTA results (**Domain-based**).
- Result:** Achieved state-of-the-art performance on multiple ADMET prediction tasks; **findings prepared for submission to a CORE-A+ Conference and then a *Nature Portfolio* journal.**

DL-based Pocket Park Planning in High-density Cities with SVI Big Data | *AI4Urban-Planning*

(SEU Undergraduate Graduation Design: Cooperation with [Prof Conghui Zhou](#))

Nov.2021 – Feb.2024

- Positioning:** The first deep learning approach based on Street View Big Data for the Pocket Park Site Selection.
- Constructed a street view information database for the main urban area of Nanjing with 458019 street view images (**Data-centric**); constructed a domain expert data annotation method based on Trueskill and Human-in-the-loop (**Domain-based & User-oriented**), which leads to the construction of a street view evaluation dataset for Pocket Park Site Selection (**Data-centric**); trained a three-angle SVI DL model for specific Pocket Park planning (**Algorithm-driven**).
- Results:** Contributed full-stack AI research and technical solutions in collaboration with three urban planning experts, **co-produced two invention patents and one JCR Q1 publication (Research visual selected as journal cover).**

Model Enhancements of Graph-based De Novo Molecule Generation | *AI4Science*

(DCAI Individual Project: Co-directed by [Prof Lei Chen](#) and MSR AI4S)

Feb.2024 – Jun.2024

- Positioning:** Improved a VAE-based SOTA model for mol-generation from a representation learning perspective.
- Built upon [MiCaM](#) by integrating dual motif segmentation schemes to enhance fragment-wise generation (**Data-Centric & Algorithm-Driven**); achieved improvements across Novelty, Uniqueness, KLDiv, and FCD in distribution-learning benchmarks.

LLM-Based Reflective Aids Agent in Multi-Source Social Media | *HCI*

(Part-time Exploratory Project of *APEX Lab*: Supervised by *Prof Mingming Fan*)

Feb.2024 – Jun.2024

- **Positioning:** Try to reorganise the human-AI interaction research system of human reflection in the Internet environment with multiple information sources .
- Participated in a literature review integrating perspectives from education and HCI (**Human-Oriented**); assisted in the iterative design and development of an LLM-based interactive prototyping system (**AI-Driven**).

ADDITIONAL PROJECT

Data Mining in A share News | *Framework Design&Full-stack Development* Oct.2023 – Dec.2023

- Details: https://github.com/Jonarck/Data-Mining-and-Knowledge-Discovery-in-A_share-News

Low Cost Military KBQA | *Framework Design&Knowledge Extraction/Fusion* Mar.2022 – Jul.2022

- Details: <https://github.com/Jonarck/Knowledge-Extraction-and-Fusion-in-Low-cost-Military-KBQA>

SKILLS

- **ENGLISH:** IELTS: 7.0(6.0) (2023.09)
- **PROGRAMMING:** Python, C/C++, Matlab, etc.
- **DL FRAMEWORK/DevOps:** PyTorch/TensorFlow/MXNet, Ubuntu, git, L^AT_EX, etc.

HOBBIES

- **Architecture & Cultural Heritage:** Exploring Over 600 National Key Cultural Heritage Sites in China
- **Reading & Poetry:** Served as the member of Yunnan Poetry Association; Established a reading club in high school;
- **Art & Speaking:** Invited to speak at TEDxSEU on how to connect with the world through my "art hobbies"